

UNIVERSITY OF THE PUNJAB

AD (2 Years) & BS (4 Years) Degree Program / First Semester – Fall 2024

Roll

Paper: Introduction to Statistics

Course Code: STAT-101



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET P

Q.1. Give short answer to the following concepts: (6x5=30)

- Prove the property:
 - $Var(c) = 0$
 - $Var(X - c) = Var(X)$, where c is a constant.
- Differentiate between following terms by giving at least two examples.
 - Descriptive and Inferential Statistics
 - Ratio and Interval Scale
- What is meant by a Histogram? Draw a histogram for the distribution of earnings (Rs.) given below:

Earnings	180 - 184	185 - 189	190 - 194	195 - 199	200 - 204	205 - 209	210 - 214
Workers	10	24	36	40	27	23	8

Following table gives the production of cotton cloth (in million meters) for Pakistan from 1954 to 1962. Find index numbers by taking average of 1958, 1959 and 1960 as base.

Year	1954	1955	1956	1957	1958	1959	1960	1961	1962
Prod.	282	389	438	470	511	555	564	630	662

The number of cars crossing a certain bridge in a big city in 10 intervals of five minutes each was recorded as follows:

25, 15, 18, 30, 35, 40, 32, 28, 18, 25, 28, 5

Calculate the arithmetic mean, median, geometric mean, mode and the harmonic mean.

The ages of 30 patients admitted in a certain hospital during a particular week are listed below. Construct a stem-and-leaf display:

48, 31, 54, 33, 18, 64, 61, 42, 40, 71, 51, 48, 52, 65, 53, 42, 39, 62, 74, 48, 29, 67, 32, 49, 68, 35, 47, 26, 42, 58

Answer the following questions.

(3x10=30)

Q2. Calculate the first four moments about the mean for the following data.

x	1	2	3	4	5	6	7	8	9
f	1	6	8	15	20	12	9	5	2

Q3. Given the following, construct Marshall-Edgeworth and Fisher's Ideal index number for 1964, taking 1957 as base year.

Commodity	1957		1964	
	Price	Quantity	Price	Quantity
Rice	9.3	100	4.5	90
Wheat	6.4	11	3.7	19
Jawar	5.1	5	2.7	3

Q4. Plot the following data as a time series. Compute 4-quarter centered moving average trend and show it on the graph.

Year	Quarters			
	1	2	3	4
2000	102	71	47	98
2001	125	106	73	231
2002	281	229	209	488
2003	484	447	457	966

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short essays on the following.

(6×5=30)

- i. Impact of Science on Society
- ii. Nucleic acid
- iii. Antibiotics
- iv. Semi-conductors
- v. Pesticides
- vi. Metabolism

Answer the following questions:

(3×10=30)

- Q.2 Write about contribution of Muslims in the development of Science?
- Q.3 Discuss Solar and lunar Day and night in detail.
- Q.4 Give a comprehensive note human behavior.

PUACP



UNIVERSITY OF THE PUNJAB

AD (2 Years) & BS (4 Years) Degree Programs / First Semester – Fall 2024

Roll No. _____

Paper: Islamic Studies / Ethics (For Non-Muslims)
Course Code: GISL-101

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

اس پرچہ کو مہیا کی گئی جوابی کاپی پر حل کریں۔

Q.1. Answer the following short questions:

(6x5=30)

سوال نمبر 1: درج ذیل سوالات کے مختصر جواب تحریر کریں۔

- Describe the five manners of reciting the Holy Quran. الف۔ قرآن مجید کی تلاوت کے پانچ آداب بیان کریں۔
- Write a blessed verse on the protection of Quran with translation. ب۔ حفاظت قرآن پر ایک آیت مبارکہ مع ترجمہ تحریر کریں۔
- Write three types of hadith. ج۔ حدیث کی تین اقسام تحریر کیجیے۔
- Write the names of the five wives of the Holy Prophet. د۔ نبی کریم ﷺ کی پانچ ازواج مطہرات کے نام لکھیے۔
- When did the Battle of Badr take place and explain its two causes? ه۔ فوج بدر کب و کہاں آپس میں لڑی؟ دو اسباب بیان کریں۔
- Write the names of Imams Ahras Figh. و۔ ائمہ اربعہ کے نام تحریر کریں۔

Answer the following questions.

(3x10=30)

درج ذیل سوالات کے تفصیلی جواب تحریر کریں۔

Q. No.2: Give translation and explanation of the following Ahadith.

(6+4=10)

سوال نمبر 2: درج ذیل حدیث کا ترجمہ و تفسیر کیجیے۔ (6+4=10)
عن أبي هريرة رضي الله عنه، عن النبي صلى الله عليه وسلم قال: اخْتَبَرُوا السَّبْعَ النُّبُوَاتِ، قَالُوا: يَا رَسُولَ اللَّهِ وَمَا هُنَّ؟ قَالَ: الشُّرُكُ بِاللَّهِ، وَالسِّحْرُ، وَقَتْلُ النَّفْسِ الَّتِي حَرَّمَ اللَّهُ إِلَّا بِالْحَقِّ، وَأَكْلُ الرِّبَا، وَأَكْلُ مَالِ الْيَتِيمِ، وَالنَّوْثِيُّ يَوْمَ الزُّحْفِ، وَقَتْلُ الْمُخَضَّاتِ النَّبَاتِ الْغَائِلَاتِ.

Q. No.3: Write a detailed note on the following.

Write a detailed note on the sermon of Hajj al-Wada.

سوال نمبر 3: درج ذیل پر تفصیلی نوٹ تحریر کیجیے۔ (10)

خطبہ حجۃ الوداع پر تفصیلی نوٹ تحریر کریں۔

Q. No.4: Write a detailed note on the following.

Write in detail on the concept of human rights in Islam.

سوال نمبر 4: درج ذیل پر تفصیلی نوٹ تحریر کیجیے۔ (10)

اسلام میں حقوق العباد کے تصور پر تفصیل سے تحریر کریں۔

ETHICS (For Non-Muslim Candidates)

Note: Answer the following questions.

(3x20=60)

نوٹ: مندرجہ ذیل سوالات کے جواب تحریر کیجیے۔

Q. 1. Explain the different concepts of Ethics (Ancient and Contemporary)

سوال نمبر 1۔ اخلاقیات کے قدیم اور عصری تصورات کی وضاحت کریں۔

Q. 2. Explain the importance of Ethics in social life.

سوال نمبر 2۔ سماجی زندگی میں اخلاقیات کی اہمیت کی وضاحت کریں۔

Q. 3. Compare the Ethical Teaching and values between Islam and Christianity.

سوال نمبر 3۔ اسلام اور عیسائیت کے درمیان اخلاقی تعلیم اور اقدار کا موازنہ کریں۔

UNIVERSITY OF THE PUNJAB

AD (2 Years) & BS (4 Years) Degree Program / First Semester – Fall 2024

Roll No. [REDACTED]

Paper: Functional English

Course Code: GENG-101

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Define the following:

(6x5=30)

- i. Define *Clipping* and *Compounding* as word formation processes, and provide two examples of each.
- ii. Explain the difference between *Complex*, and *Compound-complex* sentences. Provide relevant examples.
- iii. In the context of sound production and pronunciation, explain the difference between *Voiced* and *Voiceless* sounds. Provide two examples for each.
- iv. What is *Subject-verb Agreement*? Explain its significance with examples.
- v. In the word formation process, what do *Affixations* do? Explain with examples.
- vi. What are some key protocols of *Active Listening*?

Q.2. Answer the following questions:

(2x15=30)

1. You are the Student Coordinator of your college university's Literary Society. The society is planning to organize a literary festival on campus, which will include book readings, writing workshops, and guest lectures by renowned authors. You need permission from the university administration to use the main auditorium and surrounding event spaces. Write a **Formal Letter** to the Incharge of Student Affairs, requesting permission to hold the event.
2. Imagine that you are working as an intern at a company, and you have been asked to prepare a guide on effective communication for new interns. The guide should cover the **Core Principles of Communication**, specifically focusing on clarity, coherence, conciseness, courteousness, correctness, etc.

UNIVERSITY OF THE PUNJAB

AD (2 Years) & BS (4 Years) Degree Program / First Semester – Fall 2024

Roll No.

Paper: Civics and Community Engagement

Course Code: GCCE-101

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

اس پرچہ کو موبیا کی گئی جوابی کاپی پر حل کریں۔

Q.1. Define the following:

(5x6=30)

سوال نمبر 1: درج ذیل کی وضاحت کریں۔

i. Citizenship

(الف) شہریت

ii. Civic Engagement

(ب) شہری مصروفیت

iii. Modern Society

(ج) جدید معاشرہ

iv. Democracy

(د) جمہوریت

v. Social Justice

(ه) سماجی انصاف

Q.2. Answer the following questions:

(3x10=30)

سوال نمبر 2: درج ذیل کے تفصیلی جوابات تحریر کریں۔

1. Define the relationship between democracy and citizenship.

(الف) جمہوریت اور شہریت کا تعلق بیان کریں۔

2. Give an Overview of fundamental rights and liberties of citizens under Constitution of Pakistan 1973.

(ب) 1973ء کے آئین پاکستان کے مطابق شہریوں کی آزادی اور بنیادی حقوق بیان کریں۔

3. What are Role of advocacy in addressing social issues?

(ج) سماجی مسائل کے حل کرنے کے لئے وکالت کا کردار بیان کریں۔

UNIVERSITY OF THE PUNJAB

AD (2 Years) & BS (4 Years) Degree Program / First Semester – Fall 2024

Roll No. 3/222

Paper: Calculus

Course Code: SMTH-101

Time: 3 Hrs. Marks: 100

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Give short answer to the following concepts:

(5x6=30)

✓1. If $y = x^3 + 2x + 3$, then find $\frac{dy}{dx}$.

✓2. Examine the function defined as $f(x) = x^3 - 6x^2 + 9x$ for extreme values.

✓3. Show that $y = x^2$ has minimum value at $x = \frac{1}{e}$, if possible; otherwise, provide a justification..

X4. Show that $\varphi(x) = x$ is continuous on $\mathbb{R}$.

✓5. Solve $\int \ln x dx$, if possible; otherwise, provide a justification.

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Q.2. Answer the following questions.

(3x10=30)

X1. Solve $\int \sin^4 x dx$, if possible; otherwise, provide a justification.

✓2. Sketch the graph of the function $y = \sin 2x$.

✓3. If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2-x}$, then find $f \circ g$ and its domain.